Probability for Machine Learning

A focused probability syllabus covering the concepts
most useful for understanding Machine Learning.
Advanced probability can be layered in later alongside ML projects.

SYLLABUS • 12 CHAPTERS

Chapter 1 — Probability Fundamentals

- Probability & Outcomes
- Sample Space
- Events
- Types of Events
- Probability of an Event
- Complement of an Event

Chapter 2 — Probability Rules

- Addition Rule
- Multiplication Rule
- Union & Intersection
- Mutually Exclusive Events
- Independent Events
- Dependent Events

Chapter 3 — Conditional Probability

- Conditional Probability
- P(A | B)
- Dependent vs Independent Events
- Chain Rule of Probability
- Law of Total Probability

Chapter 4 — Bayes' Theorem

- Bayes' Theorem
- Prior Probability
- Likelihood
- Evidence
- Posterior Probability
- Bayesian Reasoning

Chapter 5 — Random Variables

- Random Variables
- Discrete Random Variables
- Continuous Random Variables
- PMF
- PDF
- CDF
- Probability Distributions

Chapter 6 — Probability Distributions

- Bernoulli Distribution
- Binomial Distribution
- Poisson Distribution
- Uniform Distribution
- Normal / Gaussian Distribution
- Exponential Distribution

Chapter 7 — Expectation & Moments

- Expected Value
- Mean of a Random Variable
- Variance
- Standard Deviation
- Moments
- Expected Value of Functions

Chapter 8 — Covariance & Correlation

- Covariance
- Positive & Negative Covariance
- Pearson Correlation
- Spearman Correlation
- Covariance Matrix

Chapter 9 — Joint & Marginal Probability

- Joint Probability
- Marginal Probability
- Joint Distributions
- Marginal Distributions
- Conditional Distributions

Chapter 10 — LLN & CLT

- Law of Large Numbers
- Central Limit Theorem (CLT)
- Sampling Means
- Convergence of Sample Averages

Chapter 11 — Sampling & Bootstrap

- Population vs Sample
- Random Sampling
- Sampling Error
- Sampling Bias
- Bootstrap Sampling
- Resampling

Chapter 12 — Probability in Machine Learning

- Probability in Classification
- Probabilistic Predictions
- Naive Bayes
- Maximum Likelihood
- MAP Estimation
- Log-Likelihood
- Prediction Uncertainty